

RSA Cryptography

1 Overview

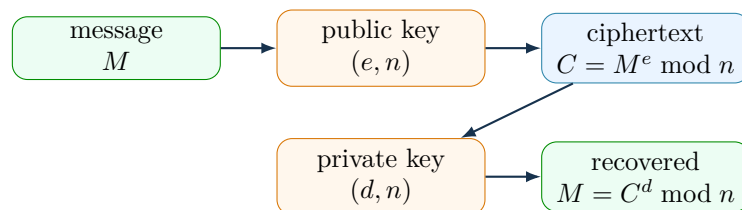
RSA

RSA is a public-key cryptosystem. It uses one key for encryption and a different related key for decryption.

public key encrypts, private key decrypts.

RSA depends on five number-theory ideas:

Idea	Role in RSA
Prime numbers	Two distinct primes p, q create the modulus $n = pq$.
Euler's totient	$\phi(n) = (p - 1)(q - 1)$ controls the exponent cycle.
Modular inverse	The private exponent d is the inverse of e modulo $\phi(n)$.
Modular exponentiation	Encryption and decryption are powers modulo n .
Factoring difficulty	Knowing n should not make it easy to recover p, q .



2 Key Generation

RSA Key Generation

1. Choose distinct primes p, q .
2. $n = pq$.
3. $\phi(n) = (p - 1)(q - 1)$.
4. Choose e with $1 < e < \phi(n)$, $\gcd(e, \phi(n)) = 1$.
5. Find d such that $de \equiv 1 \pmod{\phi(n)}$.

The public and private keys are

Public key = (e, n)

and

Private key = (d, n) .

The exponent e is public. The exponent d must remain private.

3 Encryption and Decryption

Let M be the message encoded as an integer satisfying

$$0 \leq M < n.$$

RSA Operations

$$C = M^e \bmod n$$

$$M = C^d \bmod n$$

Here C is the ciphertext. Decryption works because d was chosen to undo exponent e modulo $\phi(n)$.

4 Small RSA Example

Problem

Generate RSA keys using

$$p = 5, \quad q = 3,$$

then encrypt and decrypt

$$M = 8.$$

Solution

Compute the modulus:

$$n = pq = 5 \cdot 3 = 15.$$

Compute the totient:

$$\phi(n) = (5 - 1)(3 - 1) = 4 \cdot 2 = 8.$$

Choose

$$e = 3$$

because

$$\gcd(3, 8) = 1.$$

Now find d :

$$3d \equiv 1 \pmod{8}.$$

Since

$$3 \cdot 3 = 9 \equiv 1 \pmod{8},$$

we get

$$d = 3.$$

Thus

$$\text{Public key} = (3, 15), \quad \text{Private key} = (3, 15).$$

Encrypt:

$$C = 8^3 \bmod 15.$$

Compute

$$8^3 = 512, \quad 512 = 15(34) + 2.$$

So

$$C = 2.$$

Decrypt:

$$M = C^d \bmod 15 = 2^3 \bmod 15 = 8.$$

Answer

The original message is recovered:

$$M = 8.$$

5 Larger Key Example

Problem

Generate RSA keys using

$$p = 11, \quad q = 13,$$

and choose $e = 7$.

Solution

Compute

$$n = 11 \cdot 13 = 143.$$

Compute the totient:

$$\phi(n) = (11 - 1)(13 - 1) = 10 \cdot 12 = 120.$$

The chosen public exponent is valid because

$$\gcd(7, 120) = 1.$$

Find d from

$$7d \equiv 1 \pmod{120}.$$

Use the Euclidean step

$$120 = 7(17) + 1.$$

Rearrange:

$$1 = 120 - 7(17).$$

The coefficient of 7 is -17 , so

$$d \equiv -17 \pmod{120}.$$

Reduce:

$$-17 + 120 = 103.$$

Therefore

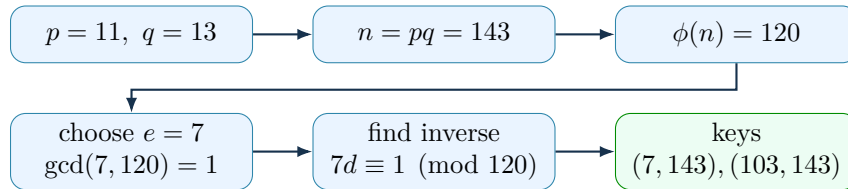
$$d = 103.$$

Answer

Public key = (7, 143)

Private key = (103, 143).

6 Key Generation Trace



7 Why Decryption Works

The private exponent d is chosen so that

$$de \equiv 1 \pmod{\phi(n)}.$$

Therefore there is an integer k such that

$$de = 1 + k\phi(n).$$

For $n = pq$,

$$\phi(n) = (p-1)(q-1),$$

so

$$de = 1 + k(p-1)(q-1).$$

During decryption,

$$C^d \equiv (M^e)^d = M^{ed} \pmod{n}.$$

We need to show

$$M^{ed} \equiv M \pmod{n}.$$

Modulo p

There are two cases.

Case 1: $M \not\equiv 0 \pmod{p}$.

By Fermat's Little Theorem,

$$M^{p-1} \equiv 1 \pmod{p}.$$

Then

$$M^{ed} = M^{1+k(p-1)(q-1)} = M \left(M^{p-1} \right)^{k(q-1)}.$$

Modulo p ,

$$M \left(M^{p-1} \right)^{k(q-1)} \equiv M(1)^{k(q-1)} \equiv M \pmod{p}.$$

Case 2: $M \equiv 0 \pmod{p}$.

Then

$$M^{ed} \equiv 0 \equiv M \pmod{p}.$$

So in all cases,

$$M^{ed} \equiv M \pmod{p}.$$

Modulo q

The same argument gives

$$M^{ed} \equiv M \pmod{q}.$$

Since p and q are coprime, the Chinese Remainder Theorem combines the two congruences:

$$M^{ed} \equiv M \pmod{pq}.$$

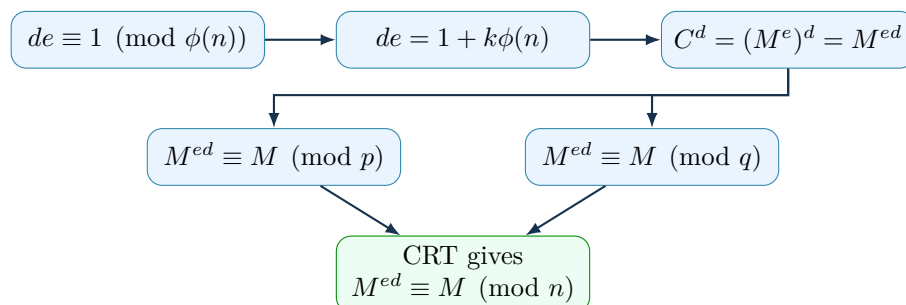
Because $n = pq$,

$$M^{ed} \equiv M \pmod{n}.$$

Conclusion

RSA decryption recovers the original message because d is chosen as the modular inverse of e modulo $\phi(n)$.

8 Correctness Flow



9 Important Notes

Toy Examples Are Not Secure

The examples in this note use tiny primes so the arithmetic is visible. Real RSA uses very large primes and also requires padding schemes in practical cryptographic systems.

Message Range

The message integer must satisfy

$$0 \leq M < n.$$

If a real message is larger than n , it must be encoded and processed in a proper block or padding scheme.

10 Review Box

Core Facts

$$\begin{aligned}n &= pq, & \phi(n) &= (p-1)(q-1). \\ \gcd(e, \phi(n)) &= 1, & de &\equiv 1 \pmod{\phi(n)}. \\ \text{Public key} &= (e, n), & \text{Private key} &= (d, n). \\ C &= M^e \bmod n, & M &= C^d \bmod n. \\ M^{ed} &\equiv M \pmod{n} \text{ explains correct decryption.}\end{aligned}$$